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Optimizing grid-tied hybrid renewable systems for EV charging in Egypt: A techno-economic analysis

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ABSTRACT

The rapid proliferation of electric vehicles (EVs) presents both an opportunity for decarbonization and a significant challenge to Egypt's power grid. This study presents a comprehensive techno-economic optimization and comparative analysis of hybrid renewable energy systems (HRES) for a high-demand EV charging plaza in three key Egyptian urban centers: Suez, 6th of October City, and Beni Suef. Utilizing HOMER Pro software, systems integrating photovoltaic (PV) panels and wind turbines were optimized for both grid-tied and standalone configurations. The results reveal a stark performance dichotomy. Optimized grid-tied configurations, comprising a single 330 kW wind turbine and 308–417 kW of PV, are exceptionally profitable, achieving a Levelized Cost of Energy (COE) as low as $0.0040/kWh. These systems operate as net energy exporters, selling over 1000,000 kWh of surplus renewable energy to the grid annually, which generates sufficient revenue to offset all capital and operational costs. In stark contrast, standalone systems require massive capital investment (Net Present Cost > $2.1 million) and oversized generation (7–8 wind turbines) to ensure energy autonomy, resulting in a COE 20–60 times higher than their grid-tied counterparts. Environmentally, the grid-tied systems provide substantial benefits, achieving a net reduction of over 700 tonnes of CO_2 emissions annually per site. This research demonstrates that grid-interactive HRES are not only economically viable but represent the optimal pathway for deploying a sustainable, profitable, and scalable EV charging infrastructure in Egypt.

1. Introduction

Electricity is vital for Egypt's economic and social advancement, since a dependable power supply fosters growth, generates employment, and underpins public services [1,2]. Nonetheless, the nation has seen recurrent blackouts and energy deficits, jeopardizing progress. To attain its objectives, Egypt requires a reliable energy infrastructure. This highlights the necessity for innovative solutions, such as hybrid renewable energy systems for electric vehicle charging, to establish a sustainable energy future [3–5]. A reliable electricity supply is essential for industrial efficiency, commercial operations, and enhancing citizens' quality of life, rendering the shift to clean energy a national imperative [6]. The global imperative to address climate change and ensure energy security has significantly accelerated the growth of the renewable energy (RE) market. Driven by increasing environmental concerns and supportive policies, the world has witnessed a substantial expansion in RE capacity. By the end of 2024, the global renewable energy capacity reached 4.448 Terawatts (TW), with 1749 GW attributed to solar energy

Abbreviations: AC, Alternating Current; Ah, Ampere-hour; CC, Cycle Charging; CFT, Temperature Correction Factor; COE, Cost of Energy; CO_2, Carbon Dioxide; CRF, Capital Recovery Factor; C_{pv}, the rated capacity of the PV array (kW).; DC, Direct Current; DOD, Depth of Discharge; D_{pv}, the PV derating factor (%).; EV, Electric Vehicle; EVCS, Electric Vehicle Charging Station; GW, Gigawatt; GWh, Gigawatt-hours; G_T, the solar radiation incident on the PV array in the current time step (kW/m^2).; $G_{T,STC}$, the incident radiation at standard test conditions (STC, typically 1 kW/m^2).; HOMER, Hybrid Optimization of Multiple Energy Resources; HRES, Hybrid Renewable Energy System; IEA, International Energy Agency; k_p, the temperature coefficient of power (%/°C).; kWh/m^2/day, Kilowatt-hours per square meter per day; LFP, Lithium Iron Phosphate; LNG, Liquefied Natural Gas; NASA, National Aeronautics and Space Administration; NPC, Net Present Cost; NREL, National Renewable Energy Laboratory; O&M, Operation and Maintenance; PHS, Pumped Hydroelectric Storage; POWER, Prediction of Worldwide Energy Resource; PV, Photovoltaic; P_{pv}, The PV array power output (kW); RE, Renewable Energy; RF, Renewable Fraction; SOC, State of Charge; STC, Standard Test Conditions; T_C, the PV cell temperature in the current time step (°C).; $T_{C,STC}$, the PV cell temperature under STC (typically 25°C).; US$, United States Dollar; V2G, Vehicle-to-Grid; WT, Wind Turbine; °C, Degrees Celsius; %/°C, Percent per degree Celsius; $/kWh, Dollars per kilowatt-hour; $/year, Dollars per year.

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and 1133 GW to wind energy [7–9]. Furthermore, the energy storage sector, crucial for integrating variable renewables, also saw significant development, with the International Energy Agency (IEA) reporting that the total volume of batteries used in the energy sector surpassed 2400 Gigawatt-hours (GWh) in 2023 [7]. This consistent and rapid growth in the RE market, including enabling technologies like battery storage, underscores a decisive global shift towards more sustainable and resilient energy systems.

Egypt's energy landscape relies heavily on fossil fuels, primarily natural gas, which made up about 79.2 % of electricity generation in 2022 [10]. Despite possessing natural gas reserves, Egypt struggles to meet rising energy demand, leading to occasional LNG imports [11]. Additionally, the transportation sector significantly contributes to fossil fuel consumption as a major consumer of petroleum products [12]. The emergence of electric vehicles (EVs) in Egypt is projected to increase the country's electricity demand [13]. The promotion of EVs has been driven by goals to reduce petroleum consumption and greenhouse gas emissions. The necessary infrastructure and regulatory frameworks remain under development [13]. The deployment of EVs is expected to raise the load on distribution networks and challenge grid stability [14,15]. Comprehensive charging infrastructure and associated grid enhancements are currently lacking, necessitating planned upgrades to accommodate the new demand [13,16]. To meet this increased load sustainably, Egypt is pursuing renewable energy integration, targeting 42 % [15,17]. Various solutions, including PV-assisted systems and hydrogen storage, have been proposed to support this growth [17,18].

While foreign investment in the EV market may stimulate economic growth, significant initial infrastructure investment is required [4]. The anticipated environmental benefits include improved air quality and reduced greenhouse gas emissions [19]. Despite the recognized need to transition to EVs for environmental and sustainability goals, several factors challenge the development of adequate charging infrastructure in Egypt. The scarcity of public charging stations remains a primary obstacle, impeding long-distance travel beyond the convenience of home charging [20,21]. High installation costs for charging posts, metering, and associated civil works constrain the establishment of a comprehensive network [22]. Furthermore, the increased demand from EV charging threatens grid stability and requires effective load management strategies to prevent potential power outages [22,23]. Addressing these challenges involves complex considerations. Optimal placement of charging stations requires balancing economic viability, technical feasibility, and environmental impact, often employing multi-criteria decision-making methods, as applied in studies like the one in Zagazig, Egypt [21]. Economic factors, including high capital requirements and the need for investment incentives, are critical [24]. Advanced techniques like digital twins for grid asset optimization [22] and strategies like public-private partnerships [25] are proposed to accelerate deployment and manage financial risks.

Integrating renewable energy sources with charging infrastructure is crucial to enhance sustainability and align with Egypt's development goals [26]. Such integration demonstrably reduces carbon emissions; one hybrid system in Egypt achieved annual reductions of 510 tons of CO_2 and 248 tons of fossil fuel use [27], while projections for other regions also indicate substantial environmental benefits [28]. Economic analyses confirm the viability, with scenarios combining grid-connected PV, storage, and other sources significantly lowering the levelized cost of energy from \$0.409/kWh to \$0.0549/kWh [29]. System performance improves when renewable sources and storage are added. A microgrid featuring solar panels, wind turbines, and energy storage achieved an average EV charging efficiency of 92 % and a peak efficiency of 94 % [30]. Integration of photovoltaics into a residential distribution network reduced system losses by 34 % to 41 % [31]. However, the inherent variability of renewable generation necessitates solutions like energy storage and optimized hybrid configurations to ensure supply consistency [32,33].

The optimal location and sizing of charging stations affect accessibility and performance. Research in Hail City identified seven stations in Zone 4 to meet forecasted demand [28]. A study in Kentucky explored using excess photovoltaic output for green hydrogen production to support EV charging [34]. Advanced optimization models have been developed to select sites and design configurations that balance energy demand, grid integration, and cost factors [35]. Studies from neighboring regions provide valuable insights into HRES applications for various energy needs, offering precedents for understanding system design complexities and local resource integration. For instance, in Libya, El-Khozondar et al. (2023) [36] designed and analyzed a standalone hybrid PV/wind/diesel system to reliably power a COVID-19 quarantine center. Their techno-economic optimization prioritized energy autonomy and cost-effectiveness for this critical healthcare facility, demonstrating the feasibility of HRES with diesel backup for uninterrupted power supply in off-grid, temporary applications. However, our research differs by focusing on permanent, grid-interactive EV charging infrastructure, which involves distinct load profiles and complex bidirectional energy flow dynamics not addressed in their standalone model.

Further Libyan research by Nassar et al. (2022) [37] presented the design and optimization of a general isolated HRES for a specific case study, aiming for energy self-sufficiency. Their work highlights methodologies for designing cost-effective and reliable standalone renewable energy systems tailored to local conditions. Our study, in contrast, explicitly compares both grid-tied and off-grid configurations, with a primary emphasis on quantifying the economic benefits of selling surplus energy back to the grid for a high-demand EV charging load, a key differentiator from purely isolated systems. Regarding energy storage, Nassar et al. (2021) [38] investigated the integration of pumped hydroelectric storage (PHS) with a hybrid PV/Wind system in a Libyan case study, focusing on dynamic performance and optimal sizing. Their dynamic system modeling showcased PHS's potential as a large-scale storage solution to enhance grid stability and reliability. While valuable for large-scale grid applications, our research focuses on Lithium Iron Phosphate (LFP) battery storage, which is more scalable and suitable for distributed EV charging stations, and we specifically analyze the economic model of direct energy sales to the grid, rather than large-scale grid stability via PHS.

Finally, while Hafez et al. (2020) [39] assessed the technical and economic feasibility of utility-scale solar energy conversion systems in Saudi Arabia, providing a regional understanding of solar potential, their focus was on large-scale power generation. Our research, conversely, is tailored to distributed HRES for EV charging infrastructure in Egypt, analyzing specific system configurations, load profiles, and the unique economic dynamics of energy exchange with the local grid. While the aforementioned studies demonstrate HRES viability, a comprehensive techno-economic comparison of grid-tied versus off-grid PV-Wind-LFP battery systems—specifically analyzing the economic implications of bidirectional grid energy exchange (station-to-grid sales) across diverse urban Egyptian settings—remains largely unexplored. This research aims to fill that gap. This study evaluates the techno-economic performance of HRES-based EVCS at three key Egyptian sites: Suez, 6th October City, and Beni Suef. Both grid-tied and off-grid configurations are modeled and optimized using HOMER Pro, focusing on minimizing COE and NPC while meeting EV charging demands and achieving high renewable penetration. A critical aspect is evaluating the economic impact of grid interaction, including energy sales to and purchases from the grid.

Numerous studies have investigated the feasibility and design of HRES-powered EVCS across various global settings to contextualize the present research within the broader landscape of sustainable EV charging solutions. Table 1 provides an overview of selected recent research, summarizing the locations, methodologies, system configurations, EV load characteristics, and key techno-economic outcomes reported. This compilation highlights the diverse approaches to address the challenge of powering EV infrastructure with renewable energy.

Table 1 reveals global efforts using various HRES configurations

Table 1
Overview of research on electric vehicle (EV) charging stations worldwide.

Ref.	Location	Method/ tool	System integration	EVCS load	Study outcomes
[40]	DR Congo	HOMER	PV/Wind/Batt	17 kWh/d, 1.4 kWp	NPC: $39,705, COE: 0.499 $/kWh,
[41]	Bangladesh	HOMER	Grid/PV/Wind/Batt	431 kWh/d, 79 kWp	NPC: $3185,840, COE: 0.424 $/kWh
[42]	Canada	HOMER	Grid/PV/Diesel/Batt	320 kWh/d	NPC: $0.835 M, COE: 0.551 $/kWh,
[43]	Korea	HOMER	Grid/PV/Wind/Batt	157.61 kWh/d, 89 kWp	NPC: $104,756, EE: 28.1 %
[44]	Pakistan	HOMER	Grid/PV/Batt	5950 kWh/d, 723 kWp	NPC: $1.75 M, COE: 0.052 $/kWh,
[45]	China	MAPSO, PSO	Grid/PV/Batt	465 kWh/d, 30 kWp	COE: 0.623 yuan/kWh
[46]	Spain	HOMER, Exp	PV/Wind/Battery	4 MWh/d, 270 kWp	Economic: (Experimental- 83.1 %),
[47]	Turkey	HOMER	PV/Wind/Batt	2.4 MWh/d, 328 kWp	NPC: $697,704, COE: 0.064 $/kWh
[48]	Germany	HOMER, PVSOL	Grid/PV/Batt	11.13 kWh/d, 2.07kWp	PV: 13,509 kWh, grid: 2338 kWh
[49]	Bangladesh	HOMER	PV/Biogas/Batt	88 kWh/d, 12.7 kWp	NPC: $56,202, CO2: 222 g/kWh,
[50]	Spain	Monte Carlo, GA	Grid/PV/Wind/Batt	379.57 kWh/d, 50 kWp	NPV: €990,445.25, IRR: 4 yr, PI: 3.533,
[51]	Romania	RETScreen	PV/EBatt/Fuelcell	223.4 kWh/d, 50 kWp	EVCS price: €16,816.4, efficiency: 90 %
[52]	Australia	MATLAB	Grid/PV/Batt	65 kWp Annual charging	cost: AUS $28,131, %
[53]	Bangladesh	HOMER	PV/Biogas/Batt	72.51 kWh/d, 13.17 kWp	COE: 0.181 $/kWh, NPC: $93,530,
[54]	India	MSSA	Grid/PV/Diesel/Batt	257 kWh/d, 25.45kWp	, NPC: $263,377
[55]	Bangladesh	MATLAB	PV/Biogas/Batt	110 kWh/d, 20kWp	Charging cost: $0.095/kWh,
[56]	India	HOMER	Grid/PV/Batt	533 kWh/d, 140 kWp	NPC: $139,484, COE: 0.042 $/kWh,
[57]	Denmark	HOMER	PV/Wind/Diesel/Batt	170 kWh/d, 12 kWp	NPC: $531,072, COE: 0.352 $/kWh, %
[58]	Morocco	HOMER	Grid/PV/Wind/Batt	102 kWp	NPC: $2.99 M, COE: 0.084 $/kWh

(PV/Wind/Battery, Grid-tied, Diesel/Biogas) and tools (HOMER, MATLAB) for EVCS, resulting in widely varying costs (NPC/COE). While grid integration is common, detailed analysis of the economic impact of selling surplus HRES energy to the grid is often limited. Furthermore, comparative analysis of grid-tied versus off-grid PV/Wind/LFP-Battery systems specifically optimized for diverse urban Egyptian contexts remains scarce. This study directly addresses these gaps, focusing on such configurations and the economics of grid interaction within Egypt, with practical implications for the future of EV infrastructure.

The primary novel contributions of this research are:

1. A comprehensive techno-economic optimization and comparison of grid-tied versus off-grid PV-Wind-LFP battery HRES specifically for EVCS across three diverse Egyptian urban settings.
2. Quantification of the economic impact of bidirectional grid energy exchange, particularly revenue from surplus energy sales, for these systems; and
3. Providing location-specific design and economic benchmarks to support sustainable EV infrastructure development in Egypt.

The primary objectives of this study are:

1. To assess the techno-economic feasibility of HRES-based EV charging stations in the selected Egyptian locations.
2. To optimize the configuration of grid-tied and off-grid PV-Wind-Battery systems for EV charging.
3. Comparing the economic performance of different system configurations based on Cost of Energy (COE) and Net Present Cost (NPC).
4. To evaluate the impact of grid interaction, including energy sales, on the economic viability of HRES-based EV charging stations.

This study is anticipated to provide actionable insights and data-driven benchmarks for policymakers, investors, and engineers involved in the planning and deployment of cost-effective and sustainable EV charging networks across Egypt, thereby supporting the nation's transition towards cleaner transportation and enhanced energy resilience.

This paper is structured as follows: Section 2 outlines the comprehensive methodology, covering site selection, resource and load assessment, and simulation tools. Section 3 details the system components, architecture, and evaluation metrics. Section 4 discusses comprehensive optimization, comparative, economic, and sensitivity analysis results. Section 5 provides the conclusion, summarizing findings and suggesting future work.

2. Methodology

2.1. Overall methodological framework

This study employs a simulation-based approach to evaluate and optimize hybrid renewable energy systems for EV charging stations. The core analysis relies on the HOMER (Hybrid Optimization of Multiple Energy Resources) Pro software, a widely recognized tool for techno-economic optimization and simulation of microgrids and distributed generation systems [59,60]. Fig. 1 depicts the complete methodological workflow, from data acquisition to final analysis.

2.2. Site selection and characteristics

For this investigation, three prominent Egyptian urban and industrial centers were selected to implement and analyze HRES-based EV

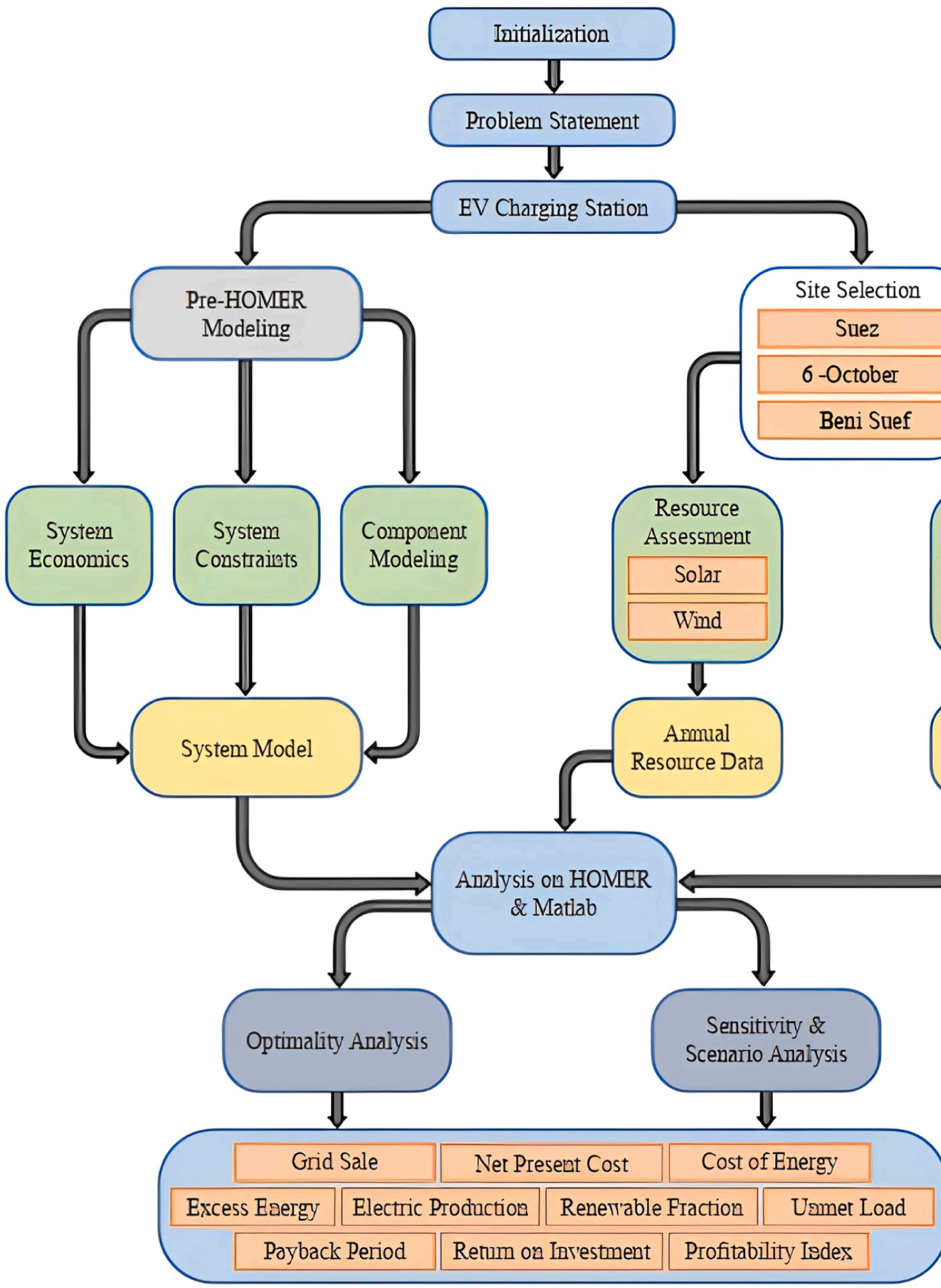

Fig. 1. The complete workflow illustration of the optimization strategy employed in this study.

charging stations: Suez, 6th October City, and Beni Suef. These locations were chosen due to their significance as major industrial hubs with substantial populations, existing grid infrastructure, and representation of different geographical regions within Egypt, potentially offering varying solar and wind resource potential. Furthermore, their roles as key transit and economic zones imply a growing demand for EV charging services. This selection is significant as it provides a comprehensive understanding of the potential for HRES-based EV charging stations in diverse urban and industrial settings within Egypt. Detailed geographical coordinates and specific characteristics of each site are provided in Table 2.

Table 2
Geographical coordinates and information of the selected areas.

Description	Study locations for EVCS		
	Suez	6-october	Beni Suef
Region	north-eastern	middle Egypt	south
Latitude (N	27°24′04.42″	46°18′42.22″	26°31′11.75″
Longitude (E)	26°29′36.71″	179°28′21.56″	26°28′17.61″
Magnitude (km2)	9002.2	1388	10,954
Elevation (m)	5	170	33

2.3. Resource assessment

Accurate assessment of local renewable energy resources is crucial for designing effective HRES. Meteorological data for the selected sites were acquired from established databases, including the National Aeronautics and Space Administration (NASA) Prediction of Worldwide Energy Resource (POWER) project, the National Renewable Energy Laboratory (NREL) database accessed via HOMER Pro, and the Egyptian Meteorological Department.

2.3.1. Solar resource

The solar energy potential at the chosen sites was assessed using monthly average daily solar irradiance (kWh/m^2/day) and clearness index data from the NREL database accessed via HOMER Pro. Fig. 2 illustrates considerable seasonal fluctuations in this data. Solar resource availability peaks during the summer (roughly May to August) across all three locations, reaching its highest point in June. In June, the peak irradiance hits about 8.4 kWh/m^2/day, coinciding with a peak clearness index of around 0.74. In contrast, solar resources are at their lowest during the winter, especially in December and January. Suez records the lowest irradiance during this time, with values falling to approximately 3.0 kWh/m^2/day in December and the lowest clearness index in January, around 0.53. Although all sites exhibit similar seasonal trends, these discrepancies underscore slight regional variations in solar potential that are significant for PV system design.

2.3.2. Wind resource

Fig. 3 illustrates the monthly average wind speed data (m/s), sourced from the NREL database via HOMER, which was used to evaluate the potential for wind energy. The data shows clear seasonal patterns and regional differences. Generally, wind speeds are greater during spring and summer (approximately April to September) than fall and winter across all three locations. Beni Suef consistently records the highest wind speeds, peaking at roughly 6.9 m/s in June. In contrast, Suez tends to have the lowest wind speeds, particularly in winter, with a minimum monthly average of about 5.2 m/s noted in January and November. 6th October City usually has wind speeds that fall between those of Beni Suef and Suez. These differences highlight varying potential for wind energy generation at the selected locations.

2.3.3. Ambient temperature

Ambient temperature data, essential for evaluating the performance of PV panels and battery systems, were sourced from the NASA surface meteorological database through HOMER Pro. Fig. 4 illustrates the three locations' monthly average temperatures (°C). A clear seasonal trend is observable, with the highest temperatures occurring in the summer months (July-August) and the lowest in winter (January). Beni Suef typically has the highest temperatures, peaking at around 30 °C in July. In contrast, Suez experiences the lowest temperatures, especially during winter, with a minimum monthly average close to 12 °C in January.

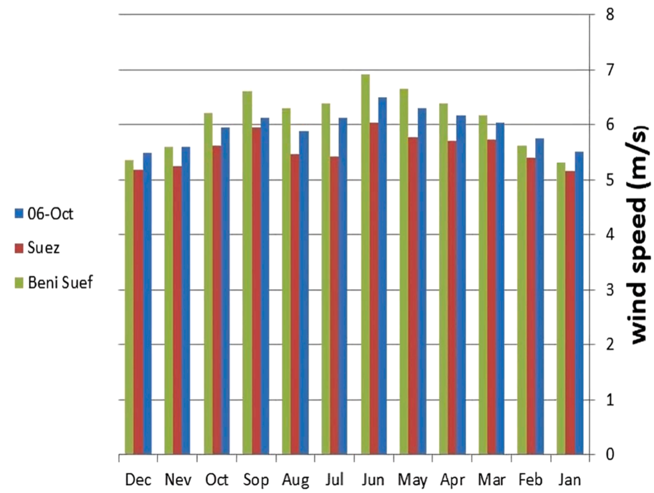

Fig. 3. Variation of wind speed in study locations.

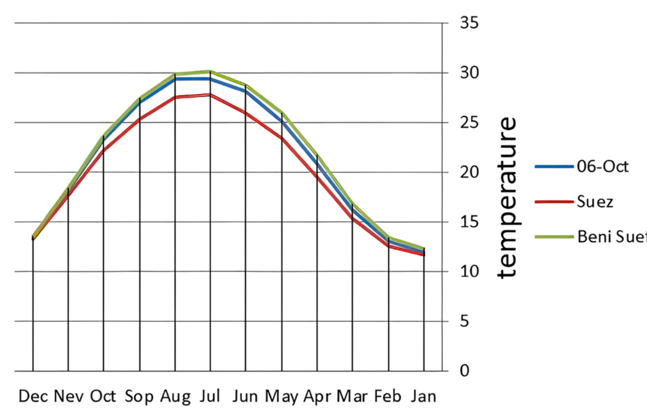

Fig. 4. Variation of temperature in study regions.

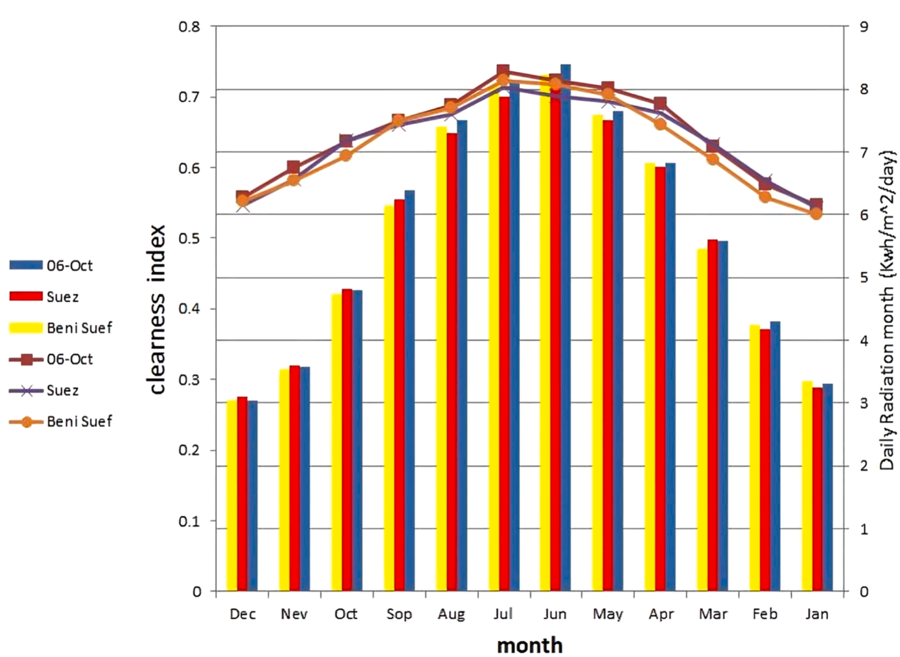

Fig. 2. Variation of clearness index and solar irradiance in study regions.

Temperatures on 6th October in the city generally fall between those of Beni Suef and Suez. These temperature patterns directly impact the efficiency calculations of components within the simulation model.

2.4. EV load profile development

To reflect a significant commercial investment, the EV charging load profile was modeled as a high traffic charging plaza designed to serve a diverse and continuous flow of electric vehicles. This scenario represents a key infrastructure node capable of supporting both private EV owners and commercial electric fleets.

The primary load characteristics, as defined in HOMER Pro, are based on this high-demand scenario:

- Average Daily Energy Demand: 2526.9 kWh/day
- Average Power Demand: 105.29 kW
- Peak Power Demand: 342 kW
- Annual Energy Demand: ~922,350 kWh/year

The load factor of 0.31 reflects a typical commercial usage pattern, with demand fluctuating throughout the day, peaking during commuter hours, and remaining active (though lower) overnight for fleet charging. This substantial load profile was used as the basis for techno-economic optimization to ensure that the designed HRES can support a viable, large-scale charging enterprise. The daily and seasonal load profiles, as depicted in Figs. 5 and 6, were scaled to match this new annual energy demand.

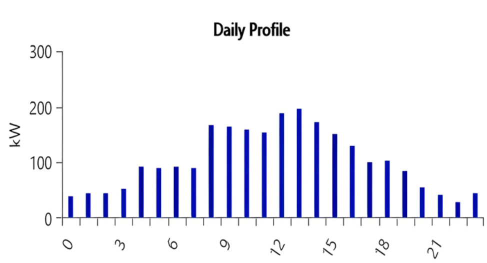

Fig. 5. Daily load profile for EV charging stations in study locations.

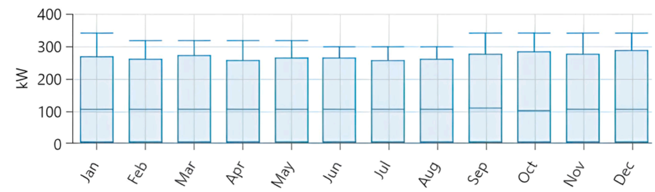

Fig. 6. Seasonal load illustration for EV charging stations.

3. System configuration and component modeling

Two primary HRES configurations were modeled and analyzed for each site: a grid-tied system and a standalone (off-grid) system. The general architecture, illustrated in Fig. 7, integrates PV panels, wind turbines (WT), LFP battery storage, and a bidirectional converter. In this configuration, the PV array and battery bank connect to the DC bus, while the wind turbine(s) and the utility grid (in the grid-tied scenario) connect to the AC bus. A bidirectional converter facilitates power flow between the DC and AC buses. The EV charging equipment is supplied by the AC bus. In the grid-tied scenario, the system can purchase electricity from the grid to meet load deficits and sell surplus renewable energy back to the grid. In the off-grid scenario, the system relies entirely on the locally generated PV and wind power, supplemented by the battery bank, to meet the EV charging demand. The subsequent subsections detail the modeling parameters for each component.

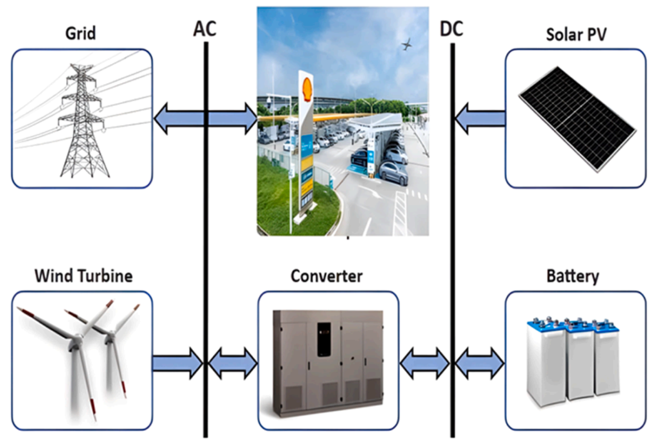

Fig. 7. System architecture of the EV charging station.

3.1. Photovoltaic (PV) panels

While solar concentrator technologies (e.g., parabolic dish systems) show potential for utility-scale applications in hot regions like Saudi Arabia and Libya [36,39], photovoltaic (PV) panels were selected for this EV charging station study. PV offers superior modularity, scalability, direct electricity generation, and established market maturity, making it more cost-effective and maintainable for distributed installations such as the EVCS considered. A 20-kW monocrystalline flat-plate PV panel with a fixed tracking system was modeled, chosen for its high efficiency, proven reliability, and widespread commercial adoption, providing a balance of performance and cost-effectiveness (technical and economic details are in Table 3). The 'Generic' manufacturer in Table 4 indicates that the model employs parameters for a typical commercially available monocrystalline flat-plate PV technology, based on industry standards and accepted performance characteristics.

The power output of the PV array (P_{PV}) at each time step was calculated by HOMER based on the available solar irradiance and the panel's cell temperature, using the following equation [61]:

$$P_{PV} = D_{PV} C_{PV} \left(\frac{G_T}{G_{T,STC}} \right) \left[1 + k_P \left(T_C - T_{C,STC} \right) \right] \quad (1)$$

In this model, the PV cell temperature (T_c) is dynamically calculated by HOMER Pro at each time step. HOMER employs a thermal model that estimates T_c based on the ambient temperature (T_a) and the incident solar radiation (G_T), utilizing the Nominal Operating Cell Temperature (*NOCT*) of the PV array [62]. This internal HOMER Pro model provides a more dynamic and accurate estimation of cell temperature than simpler linear equations by accounting for both ambient temperature and

Table 3
Technical aspects and economic details of the selected PV panel [63].

PV features	Particularization
Manufacturer	Generic
Rated Capacity (kw)	20
Panel type	Flat plate
Derating factor	80 %
Ground reflectance	20 %
Efficiency	18 %
Nominal operating temperature	47 °C
Temperature coefficient	−0.5 %/ °C
Capital cost	860 $/kW
Replacement cost	860 $/kW
O&M cost	20 $/kW/year
Lifetime	25 years

Table 4
Technical aspects and economic details of the selected wind turbine [4].

Wind turbine features	Particularization
Manufacturer	XANT l-33
Rated capacity	330 kW
Hub height	55 m
Capital cost	50,000 $
Replacement cost	50,600 $
O&M cost	500 $/year
Lifetime	25 years

incident irradiance.

3.2. Wind turbines (WT)

A generic 330 kW wind turbine (specifications in Table 4) was modeled. The capital, replacement, and O&M costs for the wind turbine are based on recent market analyses and established literature values for similar wind deployments. HOMER Pro calculates the instantaneous power output of the wind turbine based on its power curve (Fig. 8). This curve directly defines the electrical output for each wind speed. This direct lookup from the power curve ensures accurate modeling of the turbine's performance under varying wind conditions, rather than relying on simplified analytical equations for energy production. The wind speed at hub height (V_{hub}) is determined from the anemometer height wind speed (Va) using the logarithmic law:

$$V_{hub} = V_a \times \left[\frac{\ln(h_{hub}/h_s)}{\ln(h_a/h_s)}\right] \tag{2}$$

Where h represents height (hub, anemometer, roughness), the power curve (Fig. 8) shows cut-in/rated/cut-out speeds of 2.75/6.5/20 m/s. Conceptually, power (P_{WT}) depends on air density (ρ), rotor area (A), efficiencies (η), and crucially, the cube of hub wind speed:

$$P_{WT} = \frac{1}{2}\rho A \eta_{gb} \eta_{gen} V_{hub}^3 \tag{3}$$

3.3. Lithium iron phosphate (LFP) battery storage

Lithium Iron Phosphate (LFP) batteries were chosen for energy storage due to their longevity and safety characteristics. Their extended cycle life, high energy density, and thermal stability make them particularly well-suited for demanding applications like EV charging stations, ensuring long-term system reliability and reduced replacement

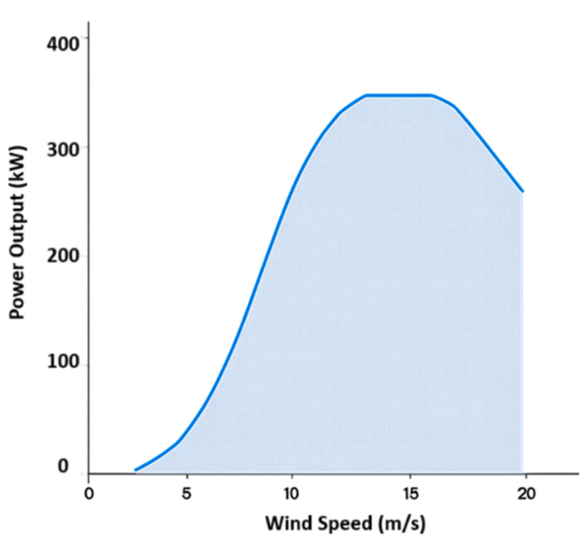

Fig. 8. The wind turbine power curve.

Table 5
Technical aspects and economic details of the selected battery [64].

Parameter	Value
Nominal Voltage (V)	12
Nominal Capacity (kWh)	16.7
Maximum Capacity (Ah)	1390
Capacity Ratio	0.297
Roundtrip Efficiency (%)	97
Capital Cost ($)	15,000.00
Replacement Cost ($)	13,800.00
O&M Cost ($/year)	1.00
Lifetime Throughput (kWh)	17,811.30
Lifetime (years)	15.00

costs. Technical specifications and costs are detailed in Table 5. These battery costs reflect current market prices for LFP technology suitable for stationary energy storage applications, derived from a synthesis of recent industry reports and academic studies [64].

The battery bank stores surplus energy generated by PV and WT and discharges to meet the load when renewable generation is insufficient. Fig. 9 illustrates the LFP battery's internal resistance and efficiency curves relative to its State of Charge (SOC). HOMER simulates the battery's energy balance at each time step. The SOC during charging ($E_{bat}(t)$) is calculated as:

$$E_{bat}(t) = E_{bat}(t-1) \times (1-\tau) + \left[E_g(t) - \frac{E_r(t)}{\eta_{inv}}\right] \times \eta_{bat} \tag{4}$$

During discharging, the required energy (Er(t)) dictates the energy drawn, accounting for efficiencies:

$$E_{bat}(t) = E_{bat}(t-1) \times (1-\tau) - \left[\frac{E_r(t)}{\eta_{inv}} - E_g(t)\right] \tag{5}$$

The total renewable energy generated ($E_g(t)$) at time t depends on the number (N) and power output ($P(t)$) of wind turbines and PV panels:

$$E_g(t) = N_{WT} \times P_{WT}(t) + N_{PV} \times P_{PV}(t) \tag{6}$$

The required capacity of the battery bank ($BESS_{cap}$) is calculated to ensure sufficient energy storage for periods of low renewable generation. The calculation incorporates the daily energy load (E_l), required autonomy days (n_{day}), battery efficiency (η_{bat}), inverter efficiency (η_{inv}), and the maximum depth of discharge (DOD). Crucially, to account for performance variations in different thermal conditions, a Temperature Correction Factor (CFT) is applied.

$$BESS_{cap} = (E_l * n_{day} * CFT)/(\eta_{bat} * \eta_{inv} * DOD) \tag{7}$$

Where the CFT is determined by the formula

$CFT = 0.0004T^2 - - 0.0246T + 1.3961$, where T is the minimum ambient temperature [65].

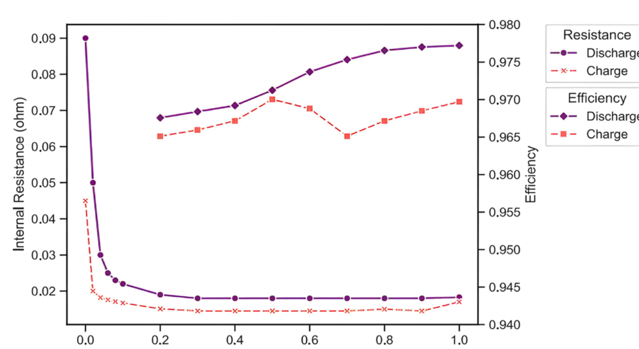

Fig. 9. Characteristic curves of the LFP battery.

3.4. Converter (Inverter/Rectifier)

A bidirectional converter manages power flow between the DC (PV, Battery) and AC (Grid, WT, Load) buses, acting as rectifiers and inverters. This type of converter is essential for enabling both grid-tied operations (allowing energy import and export) and efficient battery management within hybrid systems, ensuring flexible power flow control. The converter, listed as 'Generic' in Table 7, is modeled with industry-standard efficiencies and operational characteristics typical of commercially available bidirectional inverters/rectifiers suitable for microgrid applications. The capital, replacement, and O&M costs for the converter are based on typical market prices for bidirectional inverters/rectifiers of this capacity, as reported in recent industry benchmarks [66].

Specifications are in Table 6. Rectifier (η_{rec}) and inverter (η_{inv}) efficiencies are defined as:

$$\eta_{\text{rec}} = \frac{\text{P}_{\text{out,rec}}}{\text{P}_{\text{in,AC}}}, \tag{8a}$$

$$\eta_{\text{inv}} = \frac{\text{P}_{\text{out,inv}}}{\text{P}_{\text{in,DC}}} \tag{8b}$$

3.5. Grid connection

The utility grid is modeled as an available source and sink for electrical energy in the grid-tied scenario. HOMER simulates the interaction based on defined parameters, including the grid purchase price ($C_{p,x}$ in \$/kWh) for buying electricity, the grid sell-back price ($C_{sb,x}$ in \$/kWh) for selling surplus HRES generation, and a potential limit on the grid sale capacity (kW). The net annual cost of energy exchanged with the grid (C_{ge}) can be calculated by summing up the costs of purchased energy ($E_{gp,x,y}$) and subtracting the revenue generated from sold energy ($E_{gp,x,y}$) across different rates (x) and months (y):

$$C_{ge} = \sum_{x}\sum_{y}\left(E_{gp,x,y} \times C_{p,x}\right) - \sum_{x}\sum_{y}\left(E_{gs,x,y} \times C_{sb,x}\right) \tag{9}$$

3.6. Simulation, optimization, and economic analysis

The main goal of the HOMER Pro simulation was to determine the best HRES configuration that reduces the total NPC over a 25-year project lifespan. The NPC indicates the present value of all expenses-capital, replacement, O&M, fuel, and grid purchases- subtracted by the present value of all revenue generated (salvage value and grid sales) during the project duration. HOMER computes the NPC for each viable system setup and organizes them by rank.

$$NPC = \frac{C_{ann,tot}}{CRF(i,l)} \tag{10}$$

Where:

- $C_{ann,tot}$ = *Total annualized cost (\$/year)*
- $CRF(i,l)$ = *Capital Recovery Factor*
- i = *Real discount rate (%)*
- l = *Project lifetime (years)*

Table 6
Technical aspects and economic details of the selected converter.

Converter features	Particularization
Manufacturer	Generic
Rectifier/Inverter efficiency	95 %
Lifetime	15 years
Capital cost	300 \$/kW
Replacement cost	300 \$/kW
O&M cost	0 \$/kW/year

The Capital Recovery Factor (CRF) is calculated as:

$$CRF(i,l) = \frac{i(1+i)^l}{(1+i)^l - 1} \tag{11}$$

From the configuration with the lowest NPC, the COE is determined. The COE represents the average cost per kilowatt-hour (\$/kWh) of useful electrical energy produced by the system (energy served to the EV load plus energy sold to the grid). It is calculated as:

$$COE = \frac{C_{ann,tot}}{E_{sl} + E_{sg}} \tag{12}$$

Where:

- COE = Levelized Cost of Energy (\$/kWh)
- E_{sl}= Total load served annually (kWh/year)
- E_{sg} = Total energy sold to the grid annually (kWh/year)

HOMER's optimization algorithm explores various combinations of component sizes (PV capacity, number of wind turbines, battery capacity, converter size) to find the system architecture that meets the load demand and other constraints (e.g., minimum renewable fraction) at the lowest NPC.

All costs and economic values presented in this study are in United States Dollars (USD) based on 2024 market prices.

3.6.1. Optimization constraints

The optimization process in HOMER was subject to several operational and economic constraints to ensure realistic and feasible system designs. Primarily, the system must reliably meet the defined EV charging electrical load. A key constraint imposed in this study was a Minimum Renewable Fraction (RF) of 75 %, mandating that at least three-quarters of the energy delivered to the load originates from renewable sources (PV and Wind). HOMER calculates the RF as:

$$RF = \left(1 - \frac{E_{non,ren}}{E_s}\right) \times 100\% \tag{13}$$

Where:

- $E_{non,ren}$ = Non-renewable electrical generation serving the load (primarily grid purchases in this study) (kWh/year)
- E_s = Total electrical load served (kWh/year)

Other implicit constraints include component operational limits (e.g., battery depth of discharge, converter capacity) and economic parameters such as project lifetime (25 years), nominal discount rate (14 %), and inflation rate (25 %) [67]. Systems that fail to meet these constraints are excluded from the optimization results.

3.6.2. Dispatch strategy

HOMER employs a dispatch strategy to determine how the system operates at each time step to meet the electrical load and manage energy storage. The Cycle Charging (CC) dispatch strategy was utilized for the battery component in this study. Under Cycle Charging, when renewable power generation exceeds the load, the excess energy is first used to charge the battery bank. Once the battery reaches its setpoint state of charge, any further excess renewable generation is curtailed or, in the grid-tied scenario, sold to the grid (up to the grid sale capacity limit). When generation is less than the load, the battery discharges to meet the deficit until it reaches its minimum state of charge, after which the grid (if available) or load shedding would occur. This strategy prioritizes utilizing renewable energy and cycling the battery effectively.

3.7. Assumptions, limitations, and uncertainties

This study utilizes HOMER Pro, a widely recognized software for techno-economic optimization and simulation of hybrid renewable energy systems and microgrids [68]. Its suitability lies in its comprehensive modeling capabilities for various renewable energy components and economic analyses. However, it is important to acknowledge certain assumptions and limitations inherent to the modeling approach, which are critical for interpreting the results and guiding future research.

3.7.1. Assumptions

- The EV charging load profiles are based on an assumed representative mix of EV types and consistent daily demand throughout the year, as detailed in Section 2.4.
- Grid electricity purchase and sale prices are assumed to be static over the 25-year project lifetime.

3.7.2. Limitations

- The study employs a static grid model, meaning dynamic grid stability issues (e.g., voltage fluctuations, frequency variations) and advanced grid services (e.g., ancillary services) are not explicitly simulated. Future research could incorporate dynamic grid impact assessments.
- While HOMER Pro's internal battery model accounts for temperature effects on self-discharge and efficiency, a specific temperature correction factor (CFT) for battery capacity sizing, as discussed in Section 3.3, is not explicitly integrated into the optimization. This could lead to a slight underestimation of the required nominal capacity in extreme cold conditions.
- The component models are 'Generic,' representing typical industry standards rather than specific manufacturer products, which may not capture all nuances of real-world equipment performance.
- The study relies on historical meteorological data from established databases (NASA, NREL) and does not incorporate real-time operational data for model validation, which is a common challenge in pre-feasibility studies.

Uncertainties:

- Future fluctuations in renewable energy resource availability beyond historical patterns.
- Unforeseen changes in component costs, particularly for emerging technologies like LFP batteries, as well as grid electricity tariffs.
- Variations in EV adoption rates and charging behaviors that could alter actual load profiles and system demand.

Despite these considerations, the methodology provides a robust framework for techno-economic optimization. Future work should prioritize detailed sensitivity analyses to further assess the system's resilience to various uncertainties.

4. Results and discussion

This study presents a comprehensive techno-economic analysis of hybrid renewable energy systems across three strategically important locations in Egypt: Suez, 6-October City, and Beni Suef. The analysis compares grid-connected and off-grid configurations to determine optimal system designs and quantify economic and environmental performance. The simulation results reveal significant disparities between the two system architectures, with grid-connected systems demonstrating superior economic viability while off-grid systems require substantial capital investment to achieve energy autonomy.

4.1. Optimal system configurations

Table 7 summarizes the optimal configurations and key economic indicators for both grid-tied and off-grid systems in all locations. Grid-tied systems consistently feature a single wind turbine (330 kW), substantial PV capacity (308–417 kW), and no battery storage, leveraging the grid for both energy balancing and revenue generation. In contrast, off-grid systems require significant oversizing: 7–8 wind turbines, similar PV capacity, and large battery banks (58–66 units) to ensure reliability and autonomy.

The pronounced difference in component sizing is a direct consequence of operational requirements. Grid-tied systems can export surplus energy and purchase from the grid during deficits, allowing for leaner design. Off-grid systems, however, must be robust enough to meet all demand independently, necessitating costly redundancy in both generation and storage.

4.2. Energy dynamics

The annual energy flows for grid-tied systems are presented in Fig. 10. Across all cities, wind energy dominates the renewable mix, accounting for 60–70 % of total production, with PV providing the remainder. Notably, all grid-tied systems act as net exporters: grid sales exceed purchases by over 700,000 kWh/year in each case.

This net-exporter profile is a direct result of the high renewable fraction and the ability to sell surplus energy, which not only supports grid stability but also generates significant revenue. Grid purchases remain stable across cities (~290,000 kWh/year), indicating effective system sizing to minimize reliance on the grid.

Off-grid systems, while achieving full renewable autonomy, must overproduce (7.8–10.1 million kWh/year) to compensate for storage losses and intermittency, reflecting the inefficiencies inherent in standalone operation.

4.3. Economic viability

The economic analysis highlights a significant difference between the viability of grid-connected and off-grid systems, which has important implications for renewable energy deployment strategies in Egypt. Grid-connected systems show exceptional economic performance, with Net Present Costs ranging from $71,611 in 6th of October City to $227,909 in Suez. The corresponding Cost of Energy (COE) values range between $0.004 and $0.0126 per kWh. These remarkably low COE values illustrate the considerable impact of revenue from grid sales, which effectively subsidizes the entire operation of the system, as demonstrated in Figs. 11 and 12.

A detailed breakdown for the Suez grid-tied system (Fig. 13) shows that while capital expenditure is substantial, grid sales revenue is

Table 7
Optimal system configurations and techno-economic outcomes for grid-tied and off-grid systems in Suez, 6-October City, and Beni Suef.

City	System Type	PV (kW)	Wind Turbines	Batteries	Converter (kW)	NPC ($)	COE ($/kWh)	Renewable Fraction (%)
Suez	**Grid-connected**	417.3	1	–	263.8	$227,909	$0.013	85.00 %
6-Oct		329.6	1	–	210.6	$71,611	$0.004	85.00 %
Beni Suef		307.9	1	–	221.7	$107,180	$0.006	85.00 %
Suez	**Off-grid**	425.8	7	66	344.4	$2342,070	$0.282	100.00 %
6-Oct		423.3	7	63	297.9	$2147,320	$0.258	100.00 %
Beni Suef		421	8	58	297	$2150,232	$0.258	100.00 %

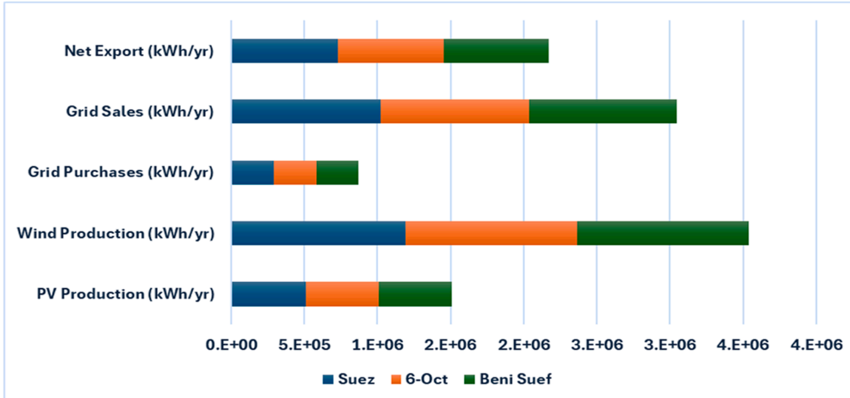

Fig. 10. Annual energy production and grid interaction for grid-tied systems.

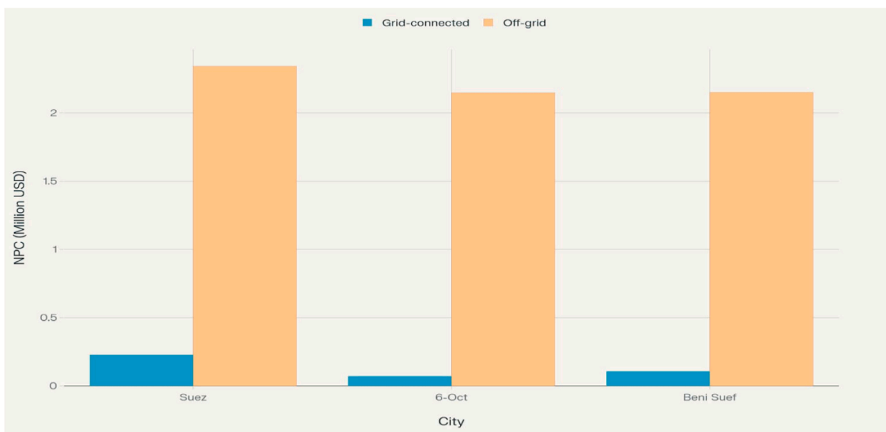

Fig. 11. The NPC between Grid-Connected and Off-Grid Systems by City.

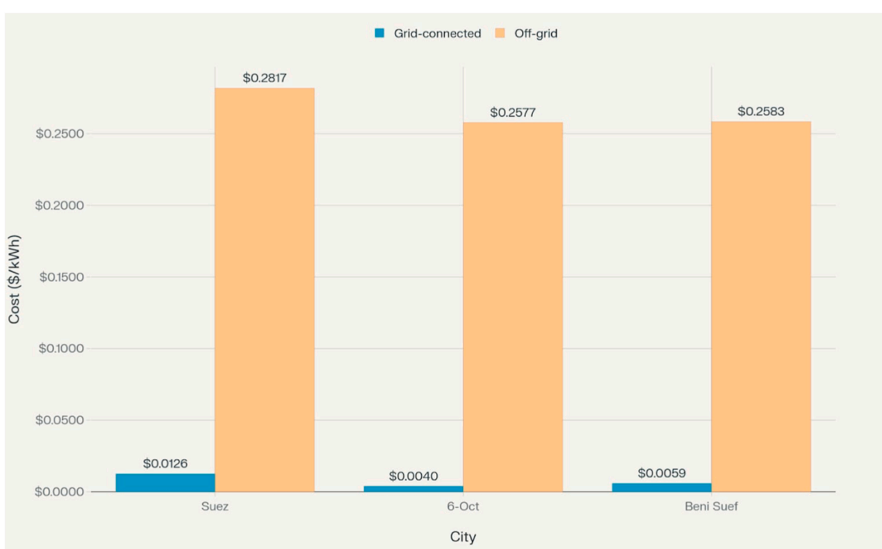

Fig. 12. The COE between Grid-Connected and Off-Grid Systems by City.

sufficient to not only cover all operating costs but also to significantly reduce the NPC. In contrast, off-grid systems lack this revenue stream and must bear the full burden of oversized generation and storage.

4.4. Environmental impact

Table 8 presents the environmental assessment for grid-tied systems. The environmental performance analysis demonstrates substantial CO_2 emission reduction benefits from grid-connected renewable energy systems across all three locations.

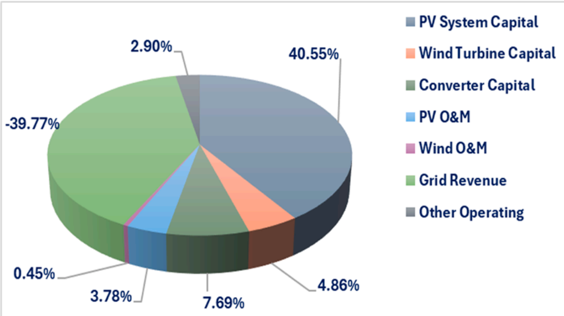

Fig. 13. Detailed breakdown of net present cost components for the Suez grid-tied system.

Table 8
Annual CO_2 emissions and net savings for grid-tied systems.

City	CO_2 from Grid Purchases (tons/yr)	CO_2 Avoided by Renewables (tons/yr)	Net CO_2 Savings (tons/yr)
Suez	151.5	883.6	732.1
6-Oct	150.9	878.1	727.2
Beni Suef	151.4	870.3	719.2

Using the Egyptian grid emission factor of 0.52 kg CO_2/kWh, the net CO_2 savings range from 719.2 tons annually in Beni Suef to 732.1 tons annually in Suez, with an average of 724.5 tons per year across all cities.

The environmental benefit calculation reveals that renewable energy production avoids 870.3–883.6 tons of CO_2 emissions annually, while grid purchases contribute only 150.9–151.5 tons of emissions. This 5:1 ratio of avoided generated emissions underscores the substantial environmental advantage of these hybrid systems. The total environmental benefit across all three cities amounts to 2173.6 tons of CO_2 savings annually, equivalent to removing approximately 470 passenger vehicles from operation.

The consistency of environmental benefits across cities, despite variations in system sizing and economic performance, reflects the dominant influence of wind generation in the renewable energy mix. Wind power's high-capacity factor (35–41 %) and substantial annual production ensure consistent environmental performance regardless of local economic optimization parameters.

5. Conclusions

This analysis demonstrates that grid-connected hybrid renewable energy systems represent the optimal solution for sustainable energy development in Egypt's strategic locations. The economic advantages are overwhelming, with grid-connected systems achieving profitability through energy sales while off-grid alternatives require massive capital investment for questionable economic returns. The environmental benefits are substantial and consistent across all locations, providing significant CO_2 emission reductions while maintaining grid stability through bidirectional power flow.

The results have important policy implications for Egypt's renewable energy strategy. Grid-connected systems can achieve rapid deployment and economic viability without subsidies, making them attractive for private investment and large-scale implementation. The net-exporter characteristic of these systems also provides grid stabilization benefits and contributes to national renewable energy targets. Off-grid systems, while technically feasible, should be reserved for remote locations where grid connection is impractical, rather than being considered as general alternatives to grid-connected solutions.

Future research should investigate the scalability of these findings across different load profiles and grid connection qualities, as well as the potential for hybrid systems to provide grid services beyond simple energy export. The integration of energy storage in grid-connected systems, while not economically justified in current analysis, may become attractive as battery costs decline and grid services markets develop.

CRediT authorship contribution statement

Mostafa M. Mousa: Writing – review & editing, Writing – original draft, Resources. **Saber M. Saleh:** Writing – review & editing, Writing – original draft, Supervision. **Mohamed M. Samy:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Investigation, Data curation. **Shimaa Barakat:** Writing – review & editing, Writing – original draft, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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